

MT209: Fundamentals of Metal Extraction

Mid-Semester Examination

Date: 17 Feb 2026

Total Marks [30]

Time: 12:00 to 13:30 hrs

1. Differentiate electro-refining with zone refining using example. [02]

2. Compare the processes route for Lead and Zinc extraction? Mention at-least one similarity and one difference. [02]

3. Two plants (A and B) performing electrolysis of same metal with same set of process parameters, including the area of electrode, volume of tanks etc. are resulting difference in production of metal. On careful examination of plant it is observed that tanks and electrodes used by Plant A were having larger width but smaller height than those used by Plant B. Comment with reason. [02]

4. The bacterial leaching process used by the industry was resulting in extremely low production rate compared to calculated ones. List the possible mistakes by the industry and suggest the remedies. [02]

5. Among Cu, Ni, Zn and Al, electrolysis is performed using fused salt for Al, and for Cu it was feasible using aqueous solution, whereas for the case of Ni and Zn the electrolysis by aqueous solution was feasible only at higher current density. Explain why? [04]

$$E^\circ_{\text{(Cu/Cu}^{2+})} = -0.337 \text{ V} \quad E^\circ_{\text{(Ni/Ni}^{2+})} = 0.25 \text{ V}$$

$$E^\circ_{\text{(Zn/Zn}^{2+})} = 0.763 \text{ V} \quad E^\circ_{\text{(Al/Al}^{3+})} = 1.66 \text{ V}$$

6. Two electro-winning plants dealing with same ore, metal, electrode, electrolyte and setup of same quality, were identified with the challenges of overpotential of different category. Both were suggested the same remedy of stirring the solution, but resulted in different results. List both category of overpotential along with the reason for difference in results. [04]

7. Given copper ore: Chalcopyrite : α %, rest is equal content of Pyrite and Silica. [04]
 Determine the % of Cu, Fe, and S in mass percent.
 α = Last two digits of your student ID

8. i) Explain the difficulty due to presence of Fe sulphides along with Cu sulphides in the copper ore, to process them through roasting followed by reduction. [01]

ii) Mention significance of flash smelting in reverberatory furnace for Cu extraction process to mitigate this issue, using reactions. [02]

iii) Explain the process of Pierce Smith Converter to obtain Cu along with reactions and diagram [02]

9. i) List the energy requirements for any hydrometallurgical process. [02]

ii) Compare the overall energy requirement with that of pyrometallurgical process, and comment which one is energy intensive and why? [03]